
Eliminating Clipping Bias in DP-SGD: Implementation and Empirical Analysis of Dice-SGD

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Abstract

Differentially Private Stochastic Gradient Descent (DP-SGD) is a widely used method for training models while preserving data privacy. It achieves this by clipping per-sample gradients, but this process introduces bias that can negatively affect model performance. Recent work by Zhang et al. (2024) proposes Dice-SGD, which uses an error-feedback mechanism to reduce this clipping bias. However, there is currently no standard implementation of this method, and its practical benefits are not well understood. In this project, we aim to implement Dice-SGD within the Opacus framework and evaluate its performance on standard datasets. We will study whether error-feedback improves convergence, reduces bias, and makes the model more robust to different hyperparameter settings compared to standard DP-SGD.

1 Motivation and Research Question

In standard DP-SGD, clipping per-sample gradients is a mathematical requirement to bound sensitivity. However, this step distorts the size and direction of the gradients, leading to biased updates known as clipping bias. While previous fixes mostly tuned clipping thresholds, Dice-SGD takes a different approach by adding an error-feedback loop: it saves the clipped residuals and adds them back during later updates. In theory, this approximates the true unclipped gradient trajectory while keeping strict privacy guarantees.

Because Dice-SGD hasn't been built into standard ML frameworks yet, it remains unclear if its theoretical benefits will actually lead to real-world improvements across different datasets and limited computational setups.

Research Question: *Does the error-feedback mechanism in Dice-SGD actually reduce clipping bias in practice, and does it make the model more robust to bad clipping norms across different privacy budgets (ϵ)?*

2 Objective, Methodology, and Experimental Setup

Our objective is to turn Dice-SGD into a deployable, fully tested optimization tool. We have structured our methodology into three phases:

1. **Implementation:** We will build Dice-SGD as a custom PyTorch optimizer natively integrated into Opacus. This involves engineering an internal state accumulator to hold clipped gradient residuals without breaking the existing privacy engine.
2. **Benchmarking:** We will test Dice-SGD against a standard DP-SGD baseline on MNIST and CIFAR-10. We will measure Test Accuracy vs. Privacy Budget (ϵ), convergence speed, and stability.

3. **Sensitivity Analysis:** We will run hyperparameter sweeps on the clipping norm (C) and analyze gradient norm distributions to empirically verify the bias reduction.

2.1 Expected Contributions & Challenges

We plan to deliver: (1) The first open-source Dice-SGD implementation within Opacus; (2) An empirical test of error feedback in DP training; and (3) Practical guidelines for deploying DP-SGD.

- **Challenge 1: Privacy Integrity.** Adding custom error accumulation risks breaking DP accounting. *Mitigation:* We will validate our code against Opacus' built-in accountant, ensuring the feedback loop only acts on post-clipped aggregates.
- **Challenge 2: Memory Overhead.** Storing residual errors requires extra memory. *Mitigation:* We will track memory usage during Colab GPU benchmarking to evaluate the trade-off between bias reduction and computational efficiency.

3 Division of Labor (Team of 2)

Tasks are strictly divided to reflect a proportional two-person effort:

- **Priyanshu Agarwal (Software Engineering):** Responsible for backend Opacus integration, translating Dice-SGD into a custom PyTorch optimizer, and managing the error accumulation logic.
- **Sudiksha Singh (Empirical Analysis):** Responsible for experimental testing, setting up MNIST/CIFAR-10 data loaders, writing training loops, running sweeps, and generating visualizations.

4 Literature Survey

1. **Differentially Private SGD Without Clipping Bias: An Error-Feedback Approach** (Zhang et al., ICLR 2024). The core reference paper detailing the mathematical formulation of Dice-SGD.
2. **Deep Learning with Differential Privacy** (Abadi et al., ACM SIGSAC 2016). Introduces the standard DP-SGD algorithm, serving as our primary baseline.
3. **Opacus: User-Friendly Differential Privacy Library in PyTorch** (Yousefpour et al., NeurIPS 2021). Mandatory for understanding the vectorization and engine abstractions for our integration.

A Declaration of LLM Usage

In accordance with the course's academic integrity policies, we disclose the following usage of Large Language Models (LLMs) during the preparation of this project proposal:

- **Did you use an LLM?** Yes.
- **For what purpose?** We used multiple LLMs (Google Gemini and OpenAI ChatGPT) as brainstorming and structural refinement tools. Gemini helped evaluate project feasibility based on hardware constraints and generated the initial LaTeX formatting. ChatGPT provided high-level structural feedback to improve the academic tone.
- **How did you verify its outputs?** All conceptual claims regarding Dice-SGD, clipping bias, and Opacus were manually verified against the course syllabus and the original Zhang et al. (2024) paper. Citations were cross-referenced via Google Scholar. The final text was manually reviewed and adapted to accurately reflect our research goals.